

Practice for lab14

Practice 14.1

Q1

← 设置

未识别的网络

按流量计费的连接

如果你的流量套餐有限制，因此想要更好地控制数据使用量，可将设备设置为按流量计费的网络。当你连接到此网络时，某些应用可能会改用节省流量模式，以减少数据使用量。

设置为流量计费的连接

☐ 关

如果设置流量上限，Windows 将为你进行按流量计费的连接设置，你不超过上限。

设置流量上限，以帮助控制在此网络上的数据使用量

IP 设置

IP 分配：手动

IPv4 地址：192.168.100.102

IPv4 子网前缀长度：24

IPv4 网关：192.168.100.1

编辑

属性

链接速度(接收/传输)：1000/1000 (Mbps)

本地链接 IPv6 地址：fe80::590:4602:2965:8f32%31

IPv6 DNS 服务器：fec0:0:0:ffff::1%1
fec0:0:0:ffff::2%1
fec0:0:0:ffff::3%1

IPv4 地址：192.168.100.102

制造商：Realtek

描述：Realtek PCIe GbE Family Controller #2

驱动程序版本：10.60.615.2022

物理地址(MAC)：00-E0-4C-9B-B4-3E

复制

正在 Ping 192.168.100.101 具有 32 字节的数据：
来自 192.168.100.102 的回复：无法访问目标主机。
来自 192.168.100.102 的回复：无法访问目标主机。
来自 192.168.100.101 的回复：字节=32 时间=47ms TTL=247
来自 192.168.100.101 的回复：字节=32 时间=48ms TTL=247

192.168.100.101 的 Ping 统计信息：
数据包：已发送 = 4，已接收 = 4，丢失 = 0 (0% 丢失)，
往返行程的估计时间(以毫秒为单位)：
最短 = 47ms，最长 = 48ms，平均 = 47ms

C:\Users\Administrator>ping 192.168.100.101

正在 Ping 192.168.100.101 具有 32 字节的数据：
来自 192.168.100.101 的回复：字节=32 时间=47ms TTL=247
来自 192.168.100.101 的回复：字节=32 时间=47ms TTL=247
来自 192.168.100.101 的回复：字节=32 时间=47ms TTL=247
来自 192.168.100.101 的回复：字节=32 时间=47ms TTL=247

192.168.100.101 的 Ping 统计信息：
数据包：已发送 = 4，已接收 = 4，丢失 = 0 (0% 丢失)，
往返行程的估计时间(以毫秒为单位)：
最短 = 47ms，最长 = 47ms，平均 = 47ms

C:\Users\Administrator>

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Q3

Method 1:

```
port link-mode route
```

Change one to route mode

Method 2:

```
mac-address blackhole 00e0-4c9b-b3a1
```

Add mac black-hole

Q4

```
[H3C]display mac-address
```

MAC ADDR	VLAN ID	STATE	PORT INDEX	AGING TIME
00e0-4c9b-b3a1	1	Learned	GigabitEthernet0/2	AGING
00e0-4c9b-b43e	1	Learned	GigabitEthernet0/3	AGING

2 items.

Dynamic.

Belong to PC.

Practice 14.3

Q1

```
[H3C]dis int g0/0
GigabitEthernet0/0 current state: UP
Line protocol current state: UP
Description: GigabitEthernet0/0 Interface
The Maximum Transmit Unit is 1500
Internet protocol processing : disabled
IP Packet Frame Type: PKTFMT_ETHNT_2, Hardware Address: 3cf5-ccbd-d319
IPv6 Packet Frame Type: PKTFMT_ETHNT_2, Hardware Address: 3cf5-ccbd-d319
Media type is twisted pair, loopback not set, promiscuous mode not set
1000Mb/s, Full-duplex, link type is autonegotiation
Output flow-control is disabled, input flow-control is disabled
output queue : (Urgent queuing : Size/Length/Discards) 0/100/0
output queue : (Protocol queuing : Size/Length/Discards) 0/500/0
output queue : (FIFO queuing : Size/Length/Discards) 0/75/0
Last clearing of counters: Never
  Last 300 seconds input rate 9.96 bytes/sec, 79 bits/sec, 0.11 packets/sec
  Last 300 seconds output rate 0.00 bytes/sec, 0 bits/sec, 0.00 packets/sec
    Input: 33 packets, 2988 bytes, 33 buffers
           0 broadcasts, 33 multicasts, 0 pauses
           0 errors, 0 runts, 0 giants
           0 crc, 0 align errors, 0 overruns
           0 dribbles, 0 drops, 0 no buffers
    output: 0 packets, 0 bytes, 0 buffers
           0 broadcasts, 0 multicasts, 0 pauses
           0 errors, 0 underruns, 0 collisions
           0 deferred, 0 lost carriers
```

Q2

Types

1. route – Layer-3 interface
2. bridge – Layer-2 switching port

```
<H3C> system-view
[H3C] interface GigabitEthernet0/0
# Convert to bridge (Layer-2) port
[H3C-GigabitEthernet-0/0] port link-mode bridge
# Convert back to route (Layer-3) port
[H3C-GigabitEthernet-/0/0] port link-mode route
```

Q3

```
[H3C]dis int brief
The brief information of interface(s) under route mode:
Link: ADM - administratively down; Stby - standby
Protocol: (s) - spoofing
Interface      Link Protocol Main IP      Description
Cellular0/0    DOWN DOWN      --
GE0/0          DOWN DOWN      --
GE0/1          DOWN DOWN      --
GE0/8          DOWN DOWN      --
GE0/9          DOWN DOWN      --
NULL0         UP    UP(s)      --
Vlan1          DOWN DOWN      192.168.1.1

The brief information of interface(s) under bridge mode:
Link: ADM - administratively down; Stby - standby
Speed or Duplex: (a)/A - auto; H - half; F - full
Type: A - access; T - trunk; H - hybrid
Interface      Link Speed  Duplex Type PVID Description
GE0/2          DOWN auto   A      A    1
GE0/3          DOWN auto   A      A    1
GE0/4          DOWN auto   A      A    1
GE0/5          DOWN auto   A      A    1
GE0/6          DOWN auto   A      A    1
GE0/7          DOWN auto   A      A    1
```

```
[H3C]int g0/2
[H3C-GigabitEthernet0/2]ip address 192.168.100.1 255.255.255.0
      ^
% Unrecognized command found at '^' position.
[H3C-GigabitEthernet0/2]
```

Directly on the physical bridge-mode interface: No.

```
[H3C]int g0/2
[H3C-GigabitEthernet0/2]port access vlan 10
[H3C-GigabitEthernet0/2]quit
[H3C]int vlan-interface10
[H3C-vlan-interface10]ip address 192.168.100.1 255.255.255.0
[H3C-vlan-interface10]
```

Indirectly via a Vlan: Yes.

Q4

```
[H3C]int g0/0
[H3C-GigabitEthernet0/0]port link-type trunk
      ^
% Unrecognized command found at '^' position.
```

No.

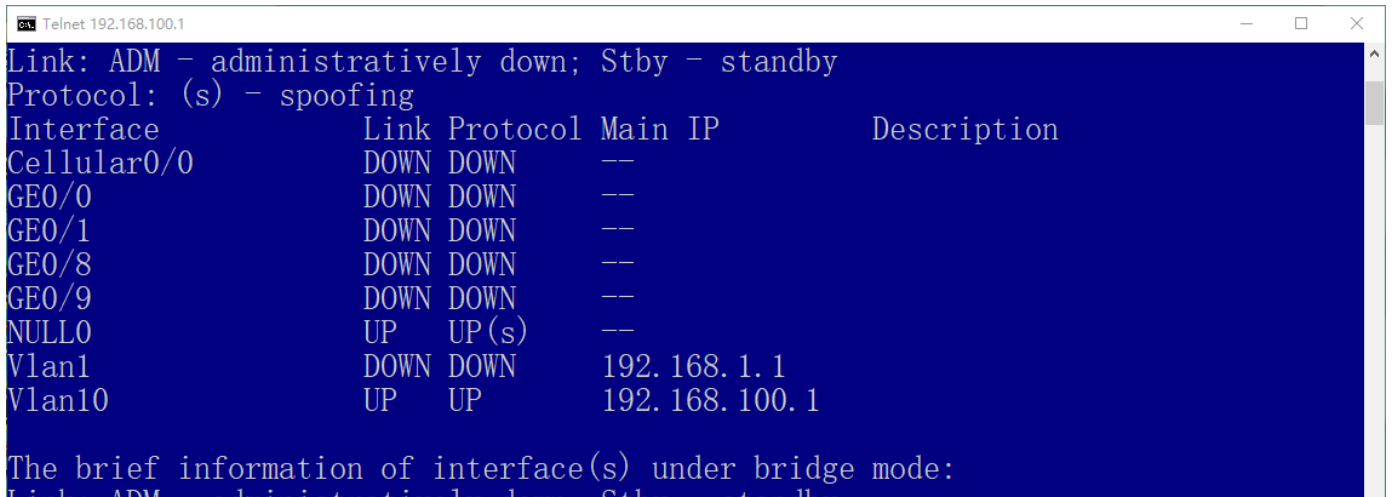
Lab Practices in Assignment 2

Q1

```
<H3C>sys
System View: return to User View with Ctrl+Z.
[H3C]dis int brief
The brief information of interface(s) under route mode:
Link: ADM - administratively down; Stby - standby
Protocol: (s) - spoofing
Interface      Link Protocol Main IP      Description
Cellular0/0    DOWN DOWN      --
GE0/0          DOWN DOWN      --
GE0/1          DOWN DOWN      --
GE0/8          DOWN DOWN      --
GE0/9          DOWN DOWN      --
NULL0          UP    UP(s)       --
Vlan1          UP    UP          192.168.1.1
Vlan10         DOWN DOWN      192.168.100.1

The brief information of interface(s) under bridge mode:
Link: ADM - administratively down; Stby - standby
Speed or Duplex: (a)/A - auto; H - half; F - full
Type: A - access; T - trunk; H - hybrid
Interface      Link Speed  Duplex Type PVID Description
GE0/2          DOWN auto   A      A    1
GE0/3          UP    1G(a)   F(a)   A    1
GE0/4          DOWN auto   A      A    1
GE0/5          DOWN auto   A      A    1
GE0/6          DOWN auto   A      A    1
GE0/7          DOWN auto   A      A    1
```

Q2



The screenshot shows a Telnet window titled "Telnet 192.168.100.1" with a dark blue background and white text. The text displays the output of the "dis int brief" command, showing the status of various network interfaces. The output is identical to the one in Q1, showing route mode interfaces (Cellular0/0, GE0/0-9, NULL0, Vlan1, Vlan10) and bridge mode interfaces (GE0/2-7).

```
Telnet 192.168.100.1
Link: ADM - administratively down; Stby - standby
Protocol: (s) - spoofing
Interface      Link Protocol Main IP      Description
Cellular0/0    DOWN DOWN      --
GE0/0          DOWN DOWN      --
GE0/1          DOWN DOWN      --
GE0/8          DOWN DOWN      --
GE0/9          DOWN DOWN      --
NULL0          UP    UP(s)       --
Vlan1          DOWN DOWN      192.168.1.1
Vlan10         UP    UP          192.168.100.1

The brief information of interface(s) under bridge mode:
Link: ADM - administratively down; Stby - standby
Speed or Duplex: (a)/A - auto; H - half; F - full
Type: A - access; T - trunk; H - hybrid
Interface      Link Speed  Duplex Type PVID Description
GE0/2          DOWN auto   A      A    1
GE0/3          UP    1G(a)   F(a)   A    1
GE0/4          DOWN auto   A      A    1
GE0/5          DOWN auto   A      A    1
GE0/6          DOWN auto   A      A    1
GE0/7          DOWN auto   A      A    1
```

```

LINK: ADM - administratively down; Stby - Standby
Speed or Duplex: (a)/A - auto; H - half; F - full
Type: A - access; T - trunk; H - hybrid
Interface      Link Speed Duplex Type PVID Description
GE0/2          DOWN auto  A    A    1
GE0/3          UP    1G(a) F(a)  A    10
GE0/4          DOWN auto  A    A    1
GE0/5          DOWN auto  A    A    1
GE0/6          DOWN auto  A    A    1
GE0/7          DOWN auto  A    A    1

[H3C]

```

Q3

Serial-COM5 - SecureCRT

File Edit View Options Transfer Script Tools Window Help

Enter host <Alt+R>

Serial-COM5 x

[RouterA]quit
 <RouterA>sys
 System View: return to user view with Ctrl+Z.
 [RouterA]int g0/0
 [RouterA-GigabitEthernet0/0]port link-mode ?
 bridge Switch to layer2 ethernet
 route Switch to layer3 ethernet
 [RouterA-GigabitEthernet0/0]port link-mode ?
 % Incomplete command found at '^' position.
 [RouterA-GigabitEthernet0/0]return
 <RouterA>sys
 System View: return to user view with Ctrl+Z.
 [RouterA]ping 192.168.100.1
 PING 192.168.100.1: 56 data bytes, press CTRL_C to break
 Reply from 192.168.100.1: bytes=56 Sequence=1 ttl=64 time=0.123 ms
 Reply from 192.168.100.1: bytes=56 Sequence=2 ttl=64 time=0.123 ms
 Reply from 192.168.100.1: bytes=56 Sequence=3 ttl=64 time=0.123 ms
 Reply from 192.168.100.1: bytes=56 Sequence=4 ttl=64 time=0.123 ms
 Reply from 192.168.100.1: bytes=56 Sequence=5 ttl=64 time=0.123 ms
 --- 192.168.100.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 1/1/1 ms
 [RouterA]
 [RouterA]
 [RouterA]
 %Jan 1 00:26:21:697 2013 RouterA IFNET/3/LINK_DOWN
 %Jan 1 00:26:21:697 2013 RouterA IFNET/5/LINEPR
 %Jan 1 00:26:21:698 2013 RouterA IFNET/3/LINK_U
 %Jan 1 00:26:21:699 2013 RouterA IFNET/3/LINK_U
 %Jan 1 00:26:21:699 2013 RouterA IFNET/5/LINEPR
 [RouterA]
 [RouterA]
 [RouterA]
 [RouterA]
 [RouterA]
 [RouterA]
 [RouterA]
 [RouterA]int g0/3
 [RouterA-GigabitEthernet0/3]port access vlan 10
 [RouterA-GigabitEthernet0/3]
 %Jan 1 00:27:14:328 2013 RouterA IFNET/3/LINK_U
 %Jan 1 00:27:14:328 2013 RouterA IFNET/5/LINEPR
 [RouterA-GigabitEthernet0/3]
 [RouterA-GigabitEthernet0/3]
 [RouterA-GigabitEthernet0/3]
 [RouterA-GigabitEthernet0/3]
 [RouterA-GigabitEthernet0/3]quit
 [RouterA]dis ip routing-table
 Routing Tables: Public
 Destinations : 4 Routes : 4
 Destination/Mask Proto Pre Cost NextHop Interface
 127.0.0.0/8 Direct 0 0 127.0.0.1 InLoop0
 127.0.0.1/32 Direct 0 0 127.0.0.1 InLoop0
 192.168.100.0/24 Direct 0 0 192.168.100.254 Vlan10
 192.168.100.254/32 Direct 0 0 127.0.0.1 InLoop0
 [RouterA]
 #Jan 1 00:27:48:851 2013 RouterA SHELL/4/LOGOUT:
 Trap 1:3.6.1.4.1.25506.2.2.1.1.3.0.2<H3CLogout>:admin logout from VTY
 %Jan 1 00:27:48:851 2013 RouterA SHELL/5/SHELL_LOGOUT: admin logged out from 192.168.1.2.█

Telnet 192.168.100.1

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 * Without the owner's prior written consent,*
 * no decompiling or reverse-engineering shall be allowed.*

<H3C>sys
 System View: return to User View with Ctrl+Z.
 [H3C]dis ip routing-table
 Routing Tables: Public
 Destinations : 4 Routes : 4
 Destination/Mask Proto Pre Cost NextHop Interface
 127.0.0.0/8 Direct 0 0 127.0.0.1 InLoop0
 127.0.0.1/32 Direct 0 0 127.0.0.1 InLoop0
 192.168.100.0/24 Direct 0 0 192.168.100.1 Vlan10
 192.168.100.1/32 Direct 0 0 127.0.0.1 InLoop0
 [H3C]